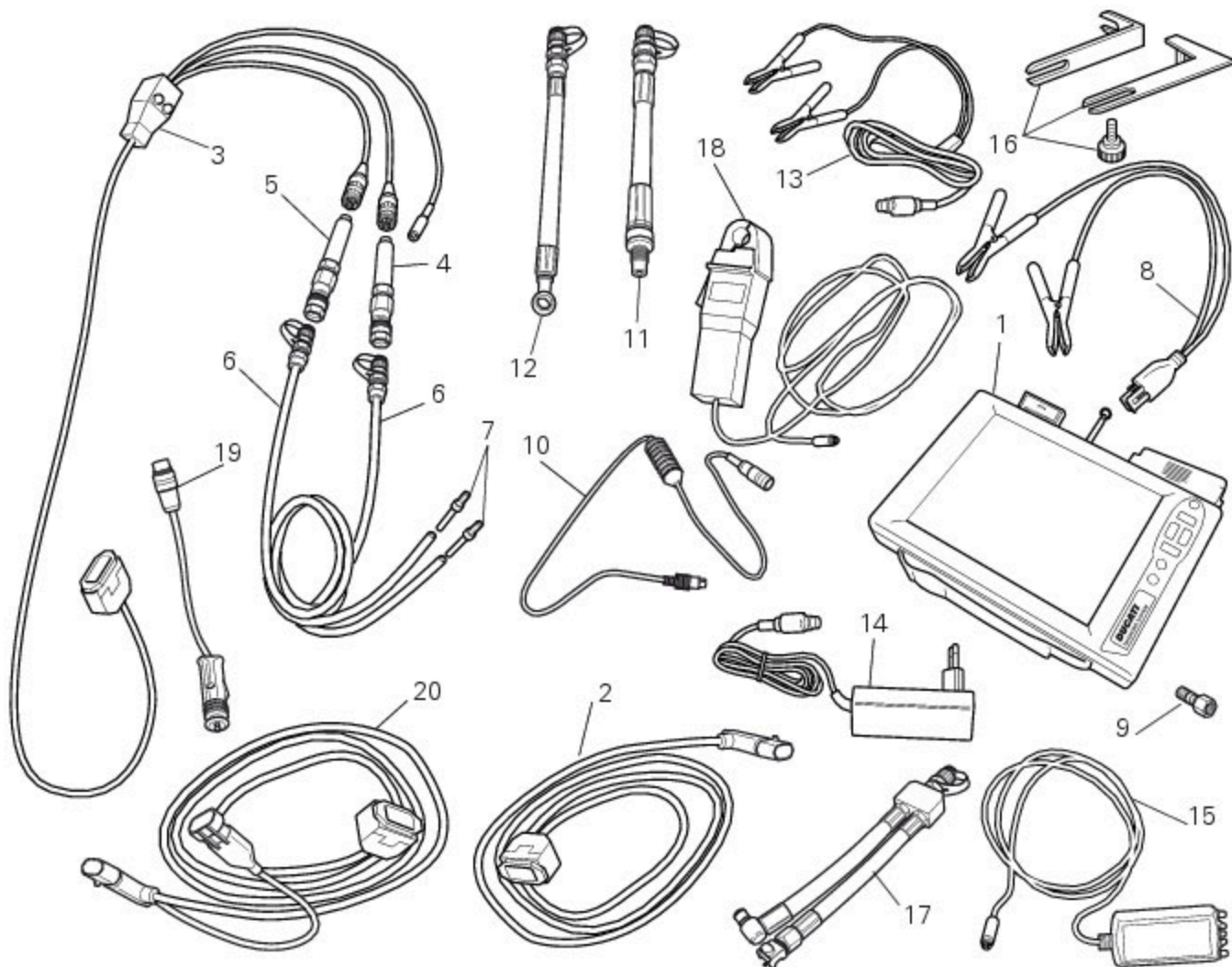


5 -Diagnostic instruments



- 1 97900.0211 Basic DDS (Ducati Diagnosis System)
- 2 97900.0227S Power and diagnostic cable SIEMENS
- 3 97900.0222 Power and diagnostic cable 1060838
(Measurement Module)
- 4 97900.0218 Vacuum sensor
- 5 552.1.039.1A Pressure sensor
- 6 97900.0220 Pressure/vacuum pipe
- 7 97900.0221 Pipe fitting
- 8 97900.0228 Battery socket adapter
- 9 814.1.114.1A Oil pressure connector
- 10 514.1.032.1A Auxiliary test cable
- 11 552.1.037.1A Cylinder compression cable with M12
connection
- 12 875.1.065.1A Oil pressure pipe
- 13 97900.0230 Mains power adapter
- 14 97900.0224 Mains power adapter
- 15 88765.1371 Belt tensioning sensor
- 16 88765.1374 Bracket for belt tensioning sensor
- 17 590.1.189.1A Fuel pressure test pipe
- 18 97900.0234 Clamp-type amperemeter
- 19 514.1.033.1A Cigar lighter socket adapter
- 20 97900.0227 ABS Power cable and diagnosis

1100 ABS [WORKSHOP SERVICE TOOLS](#)
1100 S ABS [DDS TESTER](#)
1100 S ABS [WORKSHOP SERVICE TOOLS](#)

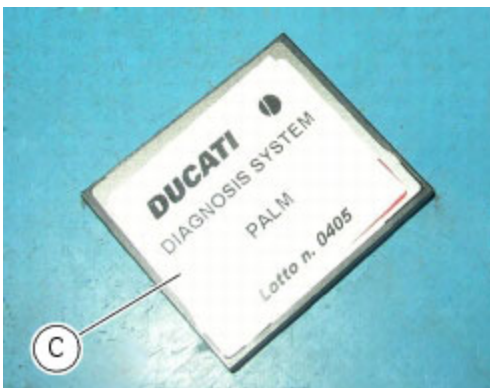
Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Description of the Ducati Diagnostic System tester

The “DDS” diagnostic system lets you diagnose any faults in the injection-ignition system via a serial port. The system is also equipped with functions to test various devices on the motorcycle. The DDS diagnosis instrument can be used to measure current and voltage on any electrical device, to perform tests on individual components and to measure pressure and temperature values.

The DDS diagnosis instrument (1) is comprised of a Palmtop display unit (A), a BBAD self-diagnosis module (B) and a display memory card (C).



The touch-screen display unit (A) serves for both data display and input, using the stylus housed on the side of the unit.

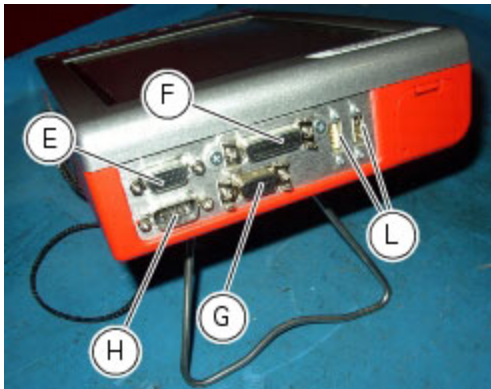
The self-diagnosis module (B) enables communication between the DDS diagnosis instrument (1) and the motorcycle's on-board electronic control unit (ECU).

The user interface software resides in the display memory card (C) which is housed in the Palmtop display unit (A).

The display unit is equipped with two connection panels: one at the top of the instrument and one at the bottom.

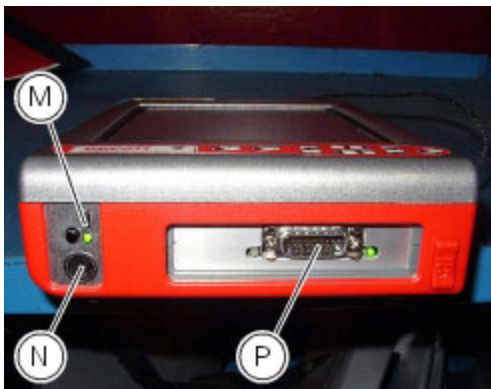
The top connection panel has 6 connection sockets with the following functions:

- one VGA output (E);
- one port for connection of the measurement module (F);
- one RS232 serial port for connection of peripheral devices (COM1) (G);
- a second RS232 serial port for connection of peripheral devices (COM2) (H);
- two generic USB ports (USB1 and USB2) (L).



The bottom connection panel has 3 connection sockets with the following functions:

- one USB port (M);
- one power connection socket (N);
- one diagnostics connection socket (P).



You can connect a printer to the DDS diagnosis instrument (1) to print test reports: connect the printer to the serial port (COM1) (G) located on the top connection panel of the tester (1).

Technical data

Power supply:

- from the mains - 220 V;
- from the motorcycle battery - 12 V.

Components supplied with the DDS diagnosis instrument

The DDS diagnosis instrument (1) is supplied in a kit together with the following items:

- Rechargeable DDS battery
- Battery charger
- Mains power adapter
- USB memory card reader
- Power and diagnostic cable complete with fuse
- CD containing DDS installation software for PC
- USB cable
- Belt tension sensor

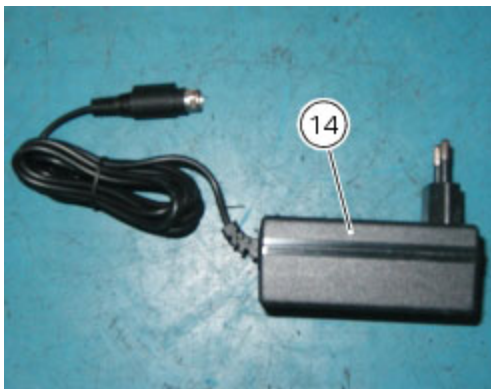
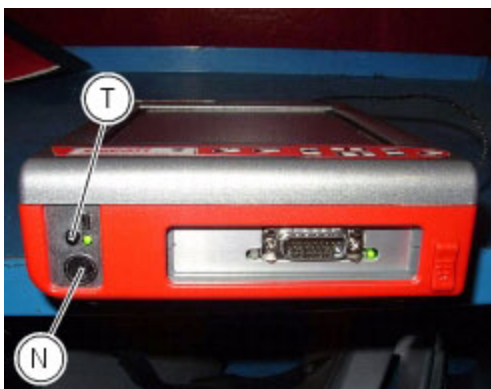
Tester power supply

The DDS diagnosis instrument (1) may be powered in any one of the following ways:



- from the mains power supply: by connecting the mains adapter (14) to the power connection socket (N);
- from the motorcycle: connecting the corresponding cables (see paragraph "[Connection to the motorcycle](#)");
- from the tester's internal battery: the battery (Q) is housed in the top part of the tester. To operate the tester (1) using the internal battery and to recharge the battery, refer to the "User Manual" supplied with the DDS diagnosis instrument.

To switch on the display, press the button (T).



Connection to the motorcycle

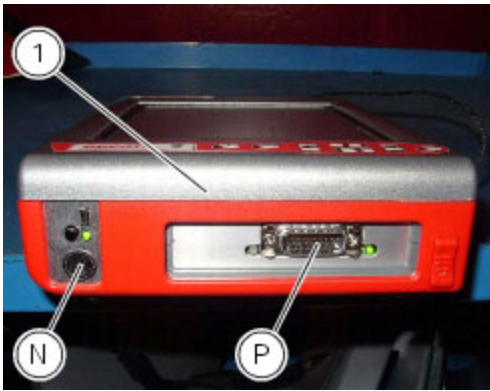
The tester requires a power supply voltage within **9** and **16** Volts D.C.

Make sure voltage never drops below this limit during operation: this situation could occur during engine starting and idling on motorcycles in which the battery is not in perfect working order. If the tester detects a power supply voltage

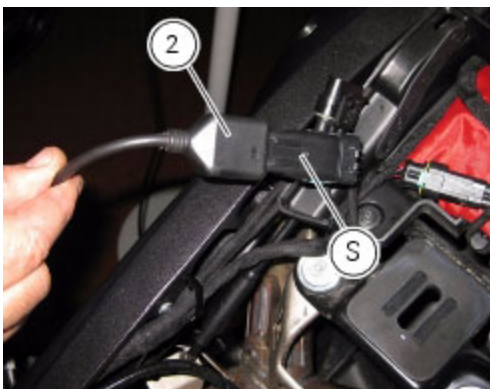
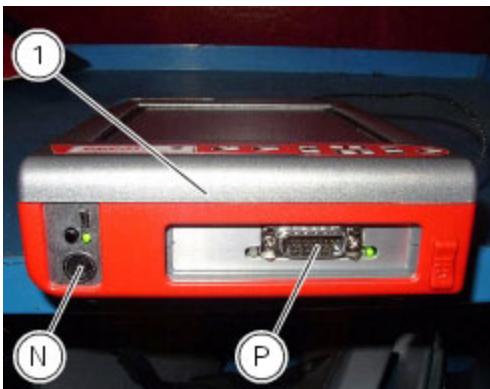
lower than **9** Volts, the current procedures will be reset; when the voltage returns to the acceptable tolerance limits the procedure must start again from the beginning.

The DDS diagnosis instrument (1) may be powered from the motorcycle in any one of the following ways:

- connect the battery power adapter (13) to the power connection socket (N) on the tester and to the motorcycle battery;



- connect the power and diagnostic cable (2) to the diagnostic connection socket (P) on the tester; then connect the motorcycle DDA socket (S) to the connection socket of the power and diagnostic cable.



For connection of the various devices required for self-diagnostic procedures, consult the instruction handbook supplied with the tester.

DDS diagnosis instrument

The main functions of the DDS diagnosis instrument can be summarised as follows:

- Retrieval of errors (faults) of the ignition-injection system stored in the engine control unit memory and their subsequent deletion, if required.
- Reading of engine parameters (rpm, coolant and air temperature, atmospheric pressure, throttle opening, battery voltage, injection times and ignition advance, etc.).
- Active diagnostics. Activation of ignition-injection system transducers to test functionality and control signals (fuel pump, ignition coils, rev counter, injectors, etc.). With this function it is also possible to enter the code to override the immobilizer.
- Road test. Allows the technician to store engine parameters recorded within a previously specified engine speed range interval. The resulting parameters can then be analysed and displayed once they have been acquired.
- With specific sensors connected, the DDS diagnosis instrument can read electrical voltages, current, temperature, timing belt tension, and pressure values (lubrication and fuel supply circuits for example).

Checking and adjusting timing belt tension



Note

The on-screen icons used during this procedure are explained in a table at the end of this section.



Note

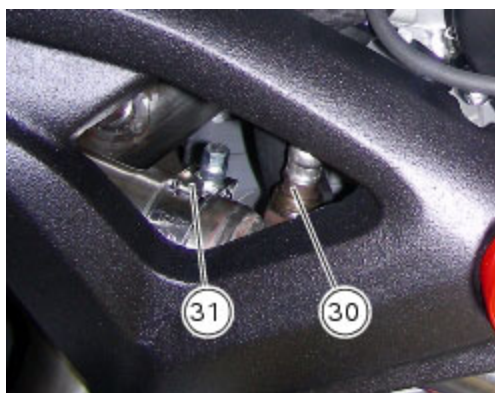
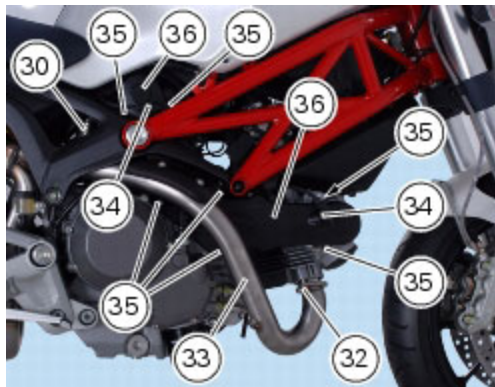
This operation, which is performed using the DDS diagnosis instrument, has the advantage that it can be carried out on both timing belts with the engine still installed on the frame. An optical reader is connected to the DDS diagnosis instrument. The optical reader has a green LED that serves to determine that the reader is correctly positioned in front of the belt to be tested. It is also equipped with an infrared transmitter (A) and receiver (B) designed to detect oscillations of the belt when caused to vibrate with the flick of a finger.

Unscrew the lambda sensor (30), undo the tie (31), loosen nuts (32) and remove the exhaust pipe (33).

Remove the spark plug caps (34).

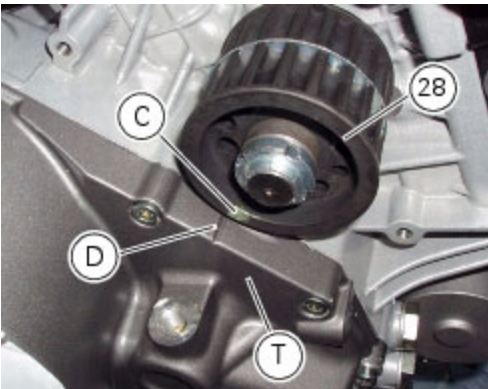
Unscrew the screws (35) and remove the side covers (36).

Remove the spark plugs.



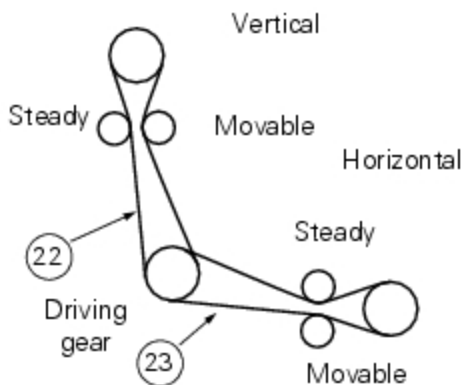


Position the crankshaft so that the piston of the horizontal cylinder is at TDC of its power stroke, using the whole tool. This is achieved by aligning the timing mark (C) of the timing driveshaft pulley (28) with the reference notch (D) on the clutch cover (T).



Measure the horizontal timing belt on the section (23) as described in paragraph "[Measuring timing belt tension](#)".

Turn the crankshaft through 270° in the direction of rotation of the engine (so that the piston of the vertical cylinder is at TDC of the power stroke) and repeat the procedure used for the horizontal cylinder, measuring the tension on section (22) of the vertical cylinder timing belt.

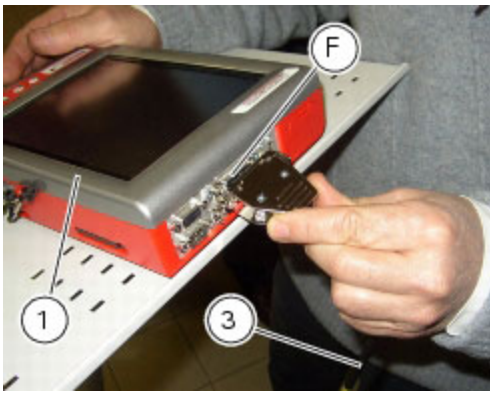


Remove the belt tension sensor and disconnect the DDS diagnosis instrument from the motorcycle. Refit all the removed parts in the reverse order to that in which they were removed.

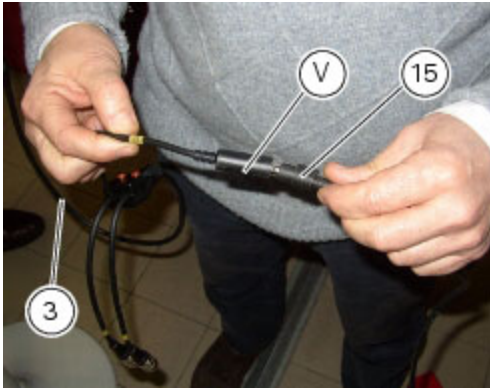
Measuring timing belt tension

Turn on the DDS diagnosis instrument (1) referring to the paragraph "[Tester power supply](#)".

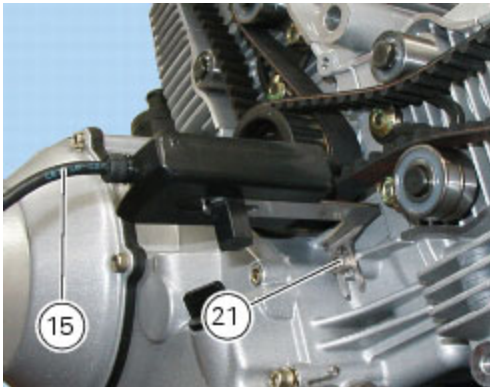
Connect the power and diagnostics cable (Measurement Module) (3) to the measurement module connector (F) on the DDS diagnosis instrument (1).



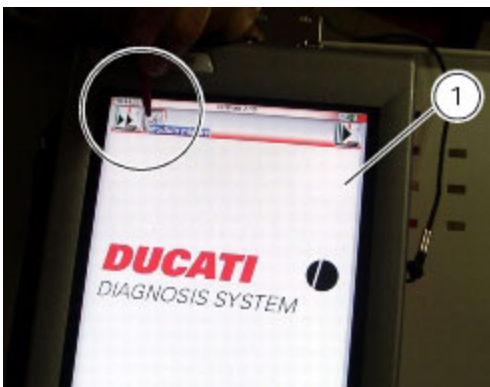
Connect the belt tension sensor (15) to the connector (V) of the power and diagnostics cable (Measurement Module) (3).

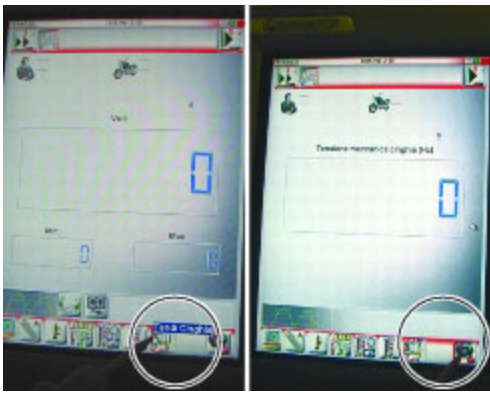


Fix the mounting bracket of the belt tension sensor (15) using the timing belt cover bolt (21). Direct the central green LED of the sensor (15) at the midpoint of the belt section, placing the sensor (15) parallel to the belt and at a distance of about **1÷1.5** cm from it.

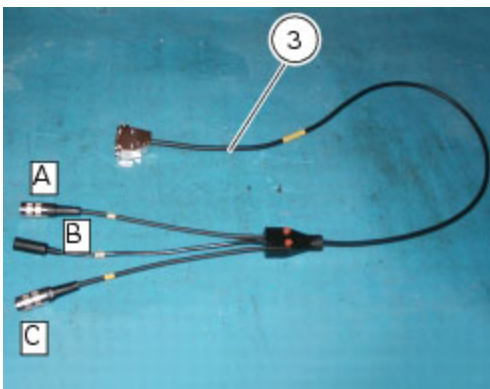


On the DDS diagnosis instrument (1), select the "Measurement module" function by pressing the corresponding icon; then press the "Belt Tension" icon followed by the "Start" icon to access the "Mechanical belt tension" screen.





The socket to which the cable (Measurement Module) (3) is to be connected is indicated on the screen with a capital letter: A, B or C.



Flick the belt lightly with your finger and read the frequency value (Hz) on the DDS diagnosis instrument.

Note

Do not stress the belt several times repeatedly, since the necessary minimum time for the DDS diagnosis instrument to take a reading is **1** second.

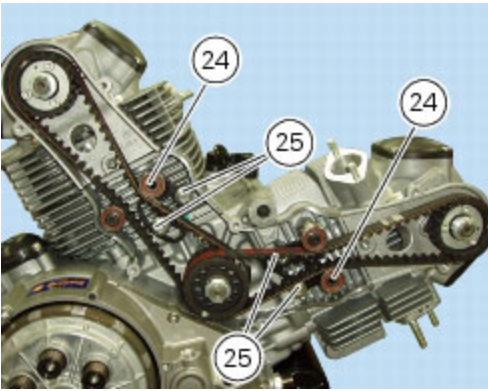
The tension values are those detailed in Sect. C 1.1, [Timing system/valves](#) and must be checked when the engine is cold: apply the installation values when fitting a new belt, and the reset values when belt tension reaches 70 Hz.

Warning

The timing belts can become slack during normal operation. When checking belt tension, if the reading is less than 70 Hz, restore to the specified nominal values (Sect. C 1.1, [Timing system/valves](#)).

If the tension value is incorrect, tighten or slacken the belt by loosening the screws (25) and moving the tensioner pulleys.

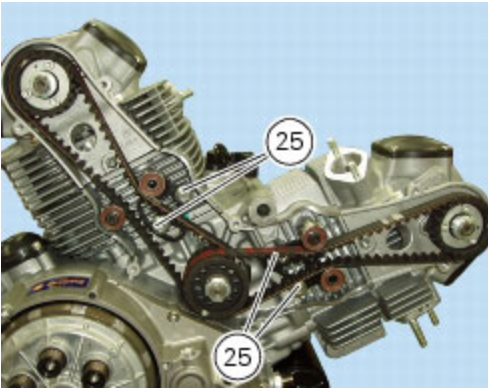
Manually raise the tensioner pulley (24) to increase belt tension or lower it to reduce belt tension.



Recheck the timing belt tension.

Repeat the above procedure until the correct belt tension is obtained.

Once belt tension is correct, tighten the screws (25) to the specified torque (Sect. C 3, [Engine torque settings](#)).



Self-adaptation of the throttle position sensor (TPS)

The linear throttle position sensor is mounted on the throttle body and cannot be adjusted. It is therefore necessary to teach the ECU the angle corresponding to the zero position of the position sensor itself, as follows.



Warning

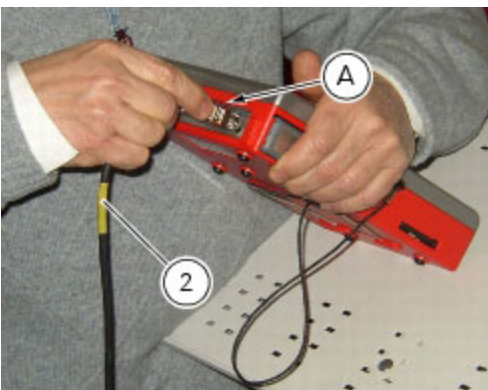
This procedure must precede the first start-up of the vehicle after replacement of the throttle body, potentiometer or ECU.

Make sure the throttle valves are seated against the stop screws (throttle twistgrip released).

In this position you can teach the ECU the angle of the position sensor. To do this, use the DDS diagnosis instrument and follow the procedure described below.

Turn on the DDS diagnosis instrument (1) referring to the paragraph "[Tester power supply](#)".

Connect the power and diagnostics cable (2) to the diagnostics connector (A) on the tester and to the diagnostics socket (B) under the seat.

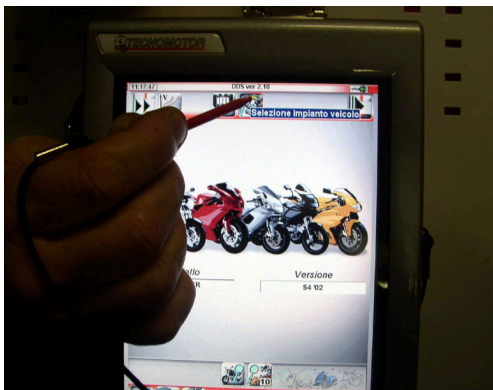




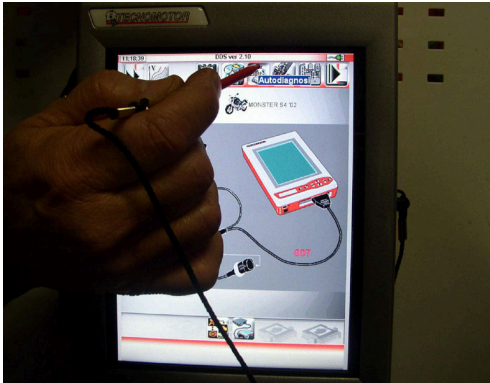
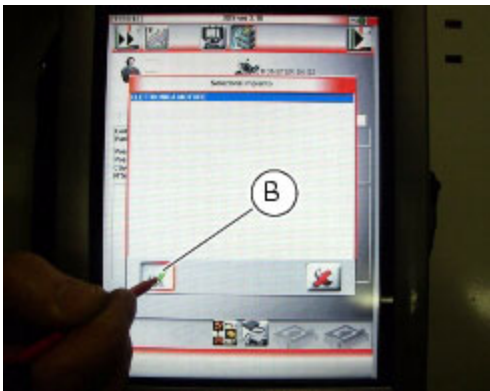
Enter the general functions menu by pressing the icon "Menu 1" (A).
 Press the "Select vehicle" icon and, on the next screen, press the "Select motorcycle" icon; select the motorcycle model and confirm, then select the version and confirm.



Press the icon "Select system" to display a list of the bike's systems that can be analysed.



Select "Engine electronics" and press the "Confirm" icon (B). Next, press the "Self-diagnosis" icon to access the corresponding function.

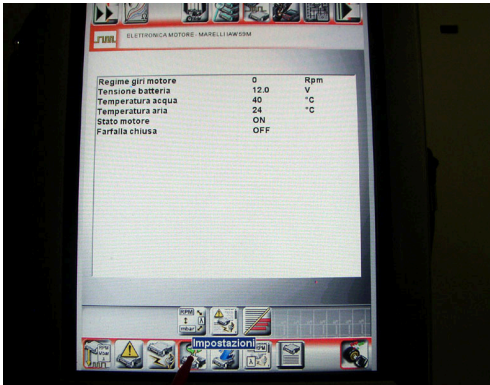


The DDS diagnosis instrument will interrogate the electronic control unit and display the parameters analysed with their relative values.

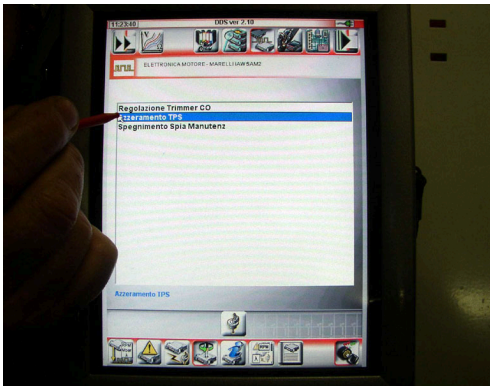


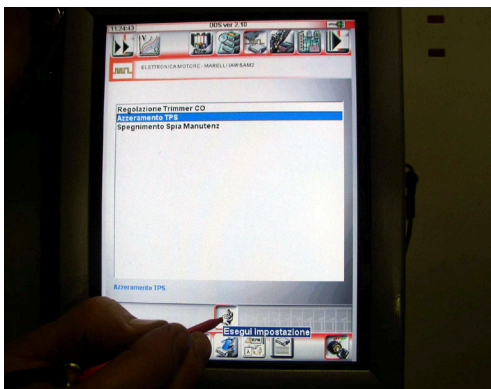
Note
Make all connections with the motorcycle ignition key in the OFF position.

Press the "Settings" icon to display the special parameters.

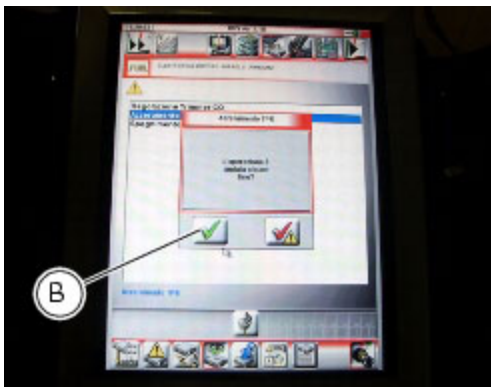


Select the option "[Self-adaptation of the throttle position sensor \(TPS\)](#)" and then press the icon "Apply".

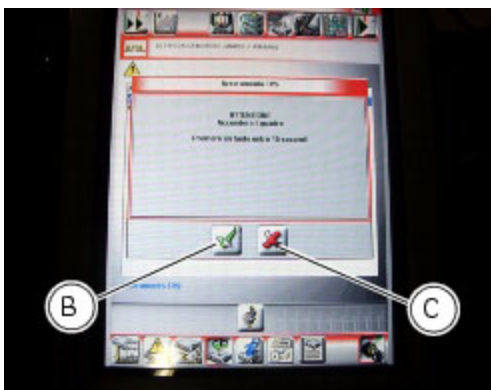




On completion of the operation, the message "Was the operation completed successfully?" will appear; press "Confirm" (B).



If any problems were encountered during the operation, the tester will display the relative error messages: each message must be confirmed (by pressing the "Confirm" icon (B)), or rejected (by pressing the "Exit" icon (C)).



Once the reset has been performed, the ECU will be associated with the motorcycle's throttle body. Replacement of the throttle body, the throttle position sensor or the ECU must always be immediately followed by a new reset procedure.

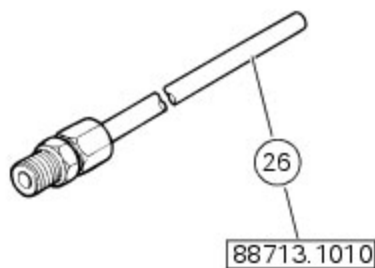
Adjusting the idle speed



Note

Check that the motorcycle is fitted with a genuine Ducati ECU, intake system and exhaust system; if not, fit genuine components.

Connect the inserts of the exhaust gas analyser code **88713.1010** to the outlets on the exhaust pipes using the fittings (26).



Connect the DDS and select MONSTER 1100 ABS.

Make sure that the throttle body pulley is seated against the stop screw when the throttle twistgrip is fully closed.

Make sure that the throttle cable free play is correctly adjusted.

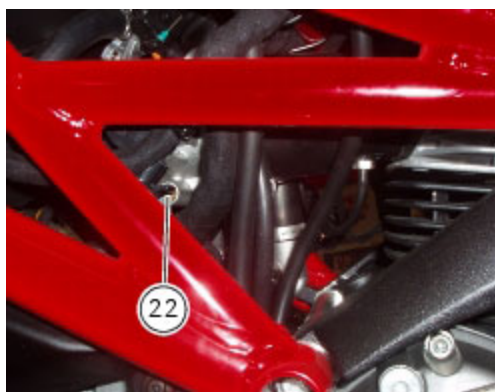
CHECK:

- that throttle angle reading with the DDS, engine off, is 0°.

Start the engine and wait for the engine temperature to reach 100 °C. When engine temperature reaches 100° C, start the following adjustment procedure.

Idle speed shall be between **1,250 and 1,350 rpm (1,300±50)**:

- If idle speed is lower or higher the specified values, open or close by 1/4 of a turn the by-pass screws (21) and (22) on both cylinders; once idle speed is fine, check that the CO rate is above 0.4% in both cylinders.



If idle speed setup procedure does not allow you to achieve the target value, proceed as follows:

- Overhaul the cylinder-head assemblies (clearance, timing, valve sealing, etc.).
- Repeat the entire idle speed adjustment procedure.
- Check the vacuum pressures downstream of the butterfly valve (carry out this check only after completing the steps described above).



Warning

If the engine is shut off before proceeding with adjustment of the CO level, wait three minutes after restarting for the enrichment phase to finish and the lambda sensors to start operating.

If idling rpm is too high and by-pass screws adjustment was not effective, make sure that parts below throttle are sealing correctly, in particular check for cracks or fissures in the intake manifolds make sure they are securely fastened to the engine, check for cuts in the pressure sensor hoses, etc.



Warning

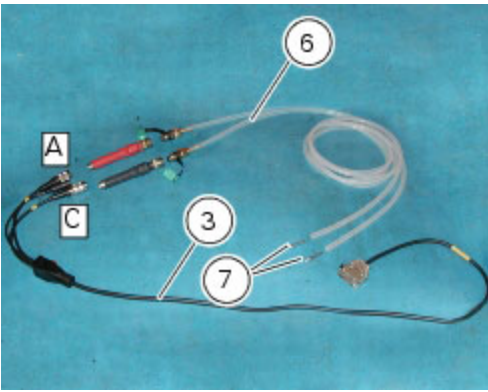
Idle speed adjustment takes priority over balancing the vacuum pressures downstream of the butterfly valve. Differences in pressure between the horizontal and vertical cylinder are therefore acceptable, provided that the idle speed is adjusted correctly.

- If you are unable to attain the desired idle speed setting, try reducing the vacuum differential downstream of the butterfly valve as described in the following paragraph "[Throttle body balancing](#)".
- Repeat the entire idle speed adjustment procedure.

Throttle body balancing

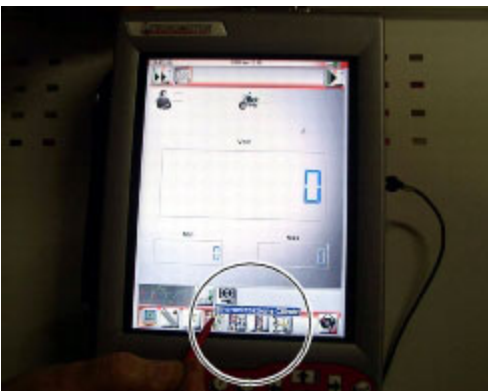
Connect the Measurement Module cable (3) to the Measurement Module connector on the DDS diagnosis instrument as described in the paragraph "[Description of the Ducati Diagnostic System tester](#)".

Connect the vacuum/pressure hoses (6) to connectors (A) and (C) of cable (3) and to unions (7).

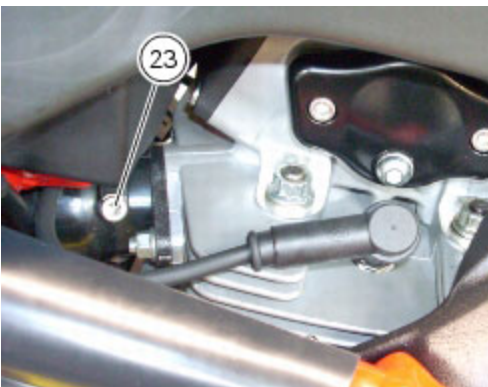


Turn on the DDS diagnosis instrument referring to the paragraph "[Tester power supply](#)". Enter the "Measurement module" function. Press the "Cylinder synchronisation" icon.

The socket to which the Measurement Module cable (3) is to be connected is indicated on the screen with a capital letter: A, B or C.

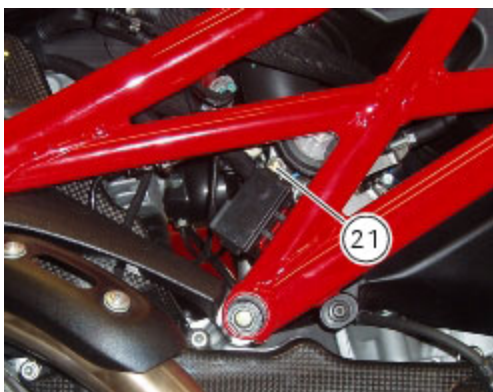


Remove the screws (23) on the intake manifolds and install the unions (7) in place of the removed screws.





Take both by-pass screws (21) and (22) one turn open, from fully closed position.



Warning

Press icon "Reset" (G) then connect the sensors (4) and (5) to connectors (A) and (C) of cable (3) and to vacuum/pressure hoses (6).

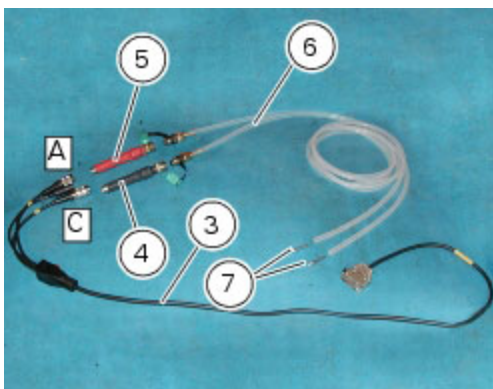
Press the "Start" icon to begin the cylinder pressure measurement procedure.

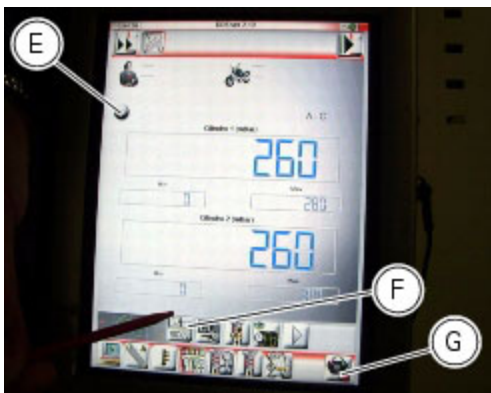
The on-screen indicator light (E) will turn red when the function is active.

Start the bike.

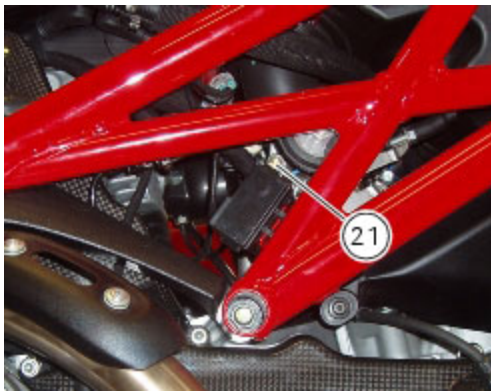
The values may be displayed in two different ways: in numerical form or in graphic form; to select the desired display type, press the "Value display" icon (F).

The values can be reset by pressing the "Reset" icon (G).





To adjust the balance (consider a tolerance between 260 and 270), turn the by-pass screws (21) and (22) until the two lines on the graph coincide, if in graphic mode, or until identical numerical values are displayed, if in numerical mode. Balance is obtained when the two lines of the graph nearly coincide or when the two indicated values are within the tolerance range.



Check the engine oil pressure

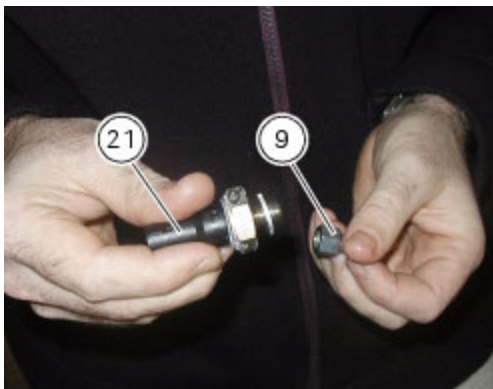


The on-screen icons used during this procedure are explained in a table at the end of this section.

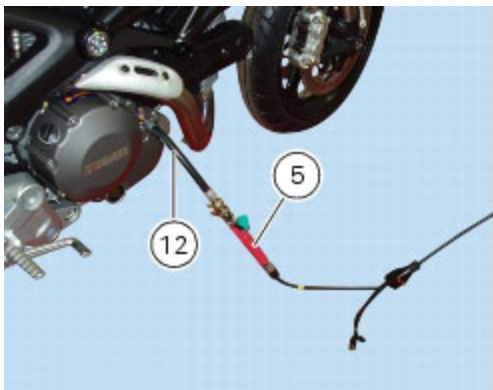
To measure the pressure of the lubrication circuit, use the engine oil pressure test point (21) as described below. Disconnect the wiring connector (A) and remove the pressure switch (21), taking care to recover the seal.



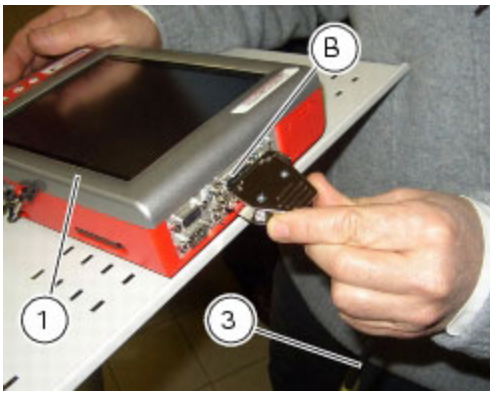
Insert the fitting (9) into the connection on the hose (12), interposing the two copper sealing washers (22). Screw the connector (9) of the hose (12) into the threaded hole (**M10x1** mm) in the RH cover. Refit the pressure switch (21) on the fitting (9) along with its seal, and reconnect the wiring connector (A).



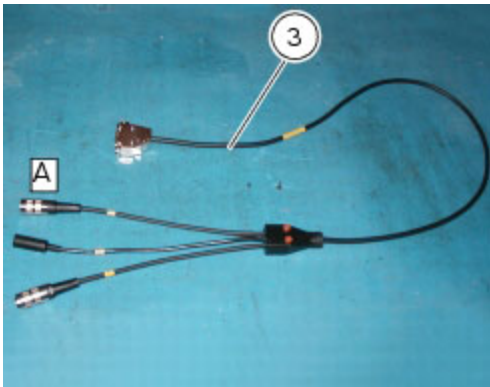
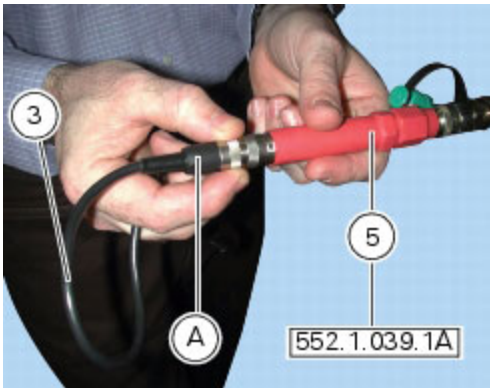
Connect the pressure sensor (5) to the pipe (12) to transform the pressure signal into an electrical signal.



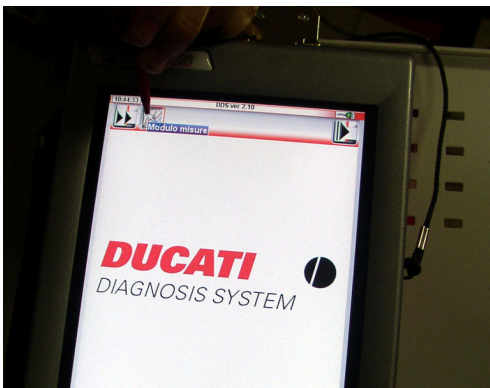
Turn on the DDS diagnosis instrument (1) referring to the paragraph "[Tester power supply](#)". Connect the power and diagnostics cable (Measurement Module) (3) to the measurement module connector (B) on the DDS diagnosis instrument (1).

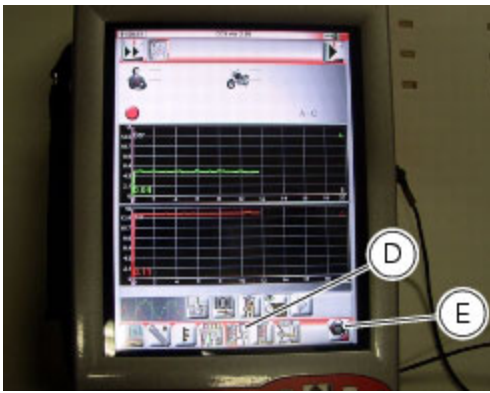


Connect the pressure sensor (5) to socket (A) of the cable (3).



On the DDS diagnosis instrument (1), select the "Measurement module" function by pressing the corresponding icon; then press the "Pressure Test" icon (D) followed by the "Start" icon (E).





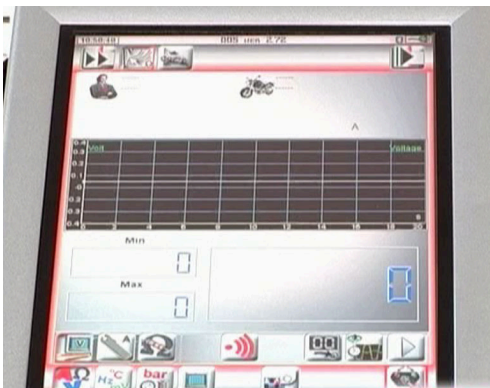
The letter A is displayed on the screen to indicate the connection socket of the Measurement Module cable (3).



Start the engine.

The values may be displayed in three different ways: in one numeric form and in two graphic forms; to select the desired display type, press the "Value display" icon.

The measured value is indicated alongside the letter (A) or (C) identifying the cable used for the measurement: i.e. if you used connector (A) of the cable (3), the value measured will be displayed next to the letter (A) on the screen.



Oil pressure test values:

Warm engine

(Minimum oil temperature = **80 °C**)

1,100 to 1,300 rpm

over **0.8 bars**

3,500 to 4,000 rpm

over **4 bars**



Important

The maximum pressure must never exceed **6.0 bar**.

Excessively high pressure may indicate that the relief valve is stuck. Excessively low pressure, on the other hand, may be caused by the relief valve being stuck in the open position, the relief valve spring being too weak, or a faulty oil pump. Other causes are excessively worn seals and gaskets or a badly worn engine.

Remove the tool and refit the pressure switch (21) along with its seal. Tighten it to the specified torque (Sect. C 3, [Engine torque settings](#)).

Reconnect the electric system connector (A) to the pressure switch.



Disconnect the DDS diagnosis instrument.

Cylinder compression test



Note

The on-screen icons used during this procedure are explained in a table at the end of this section.

Engine performance is directly correlated to the pressure that can be measured in the combustion chambers of the two cylinders. Pressure which is too high/low or an excessive difference between the two cylinders will cause a drop in engine performance and can cause engine breakdowns.

Run the engine so that it warms up to the point that the fan is tripped at least once.

Remove the HT leads (20) from both spark plugs.

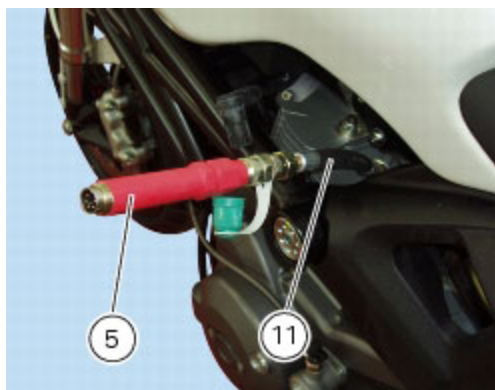
Remove the spark plug from the cylinder to be tested.



Position the HT lead carefully to avoid sparking.

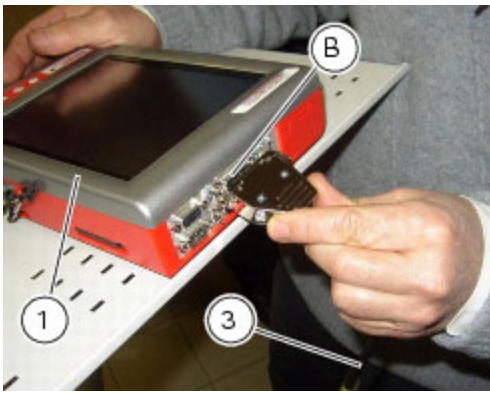
Screw the cylinder compression fitting (11) into the spark plug bore.

Connect the pressure sensor (5) to the cable (11).



Turn on the DDS diagnosis instrument (1) referring to the paragraph ["Connection to the motorcycle"](#).

Connect the power and diagnostics cable (Measurement Module) (3) to the measurement module connector (B) on the DDS diagnosis instrument (1).

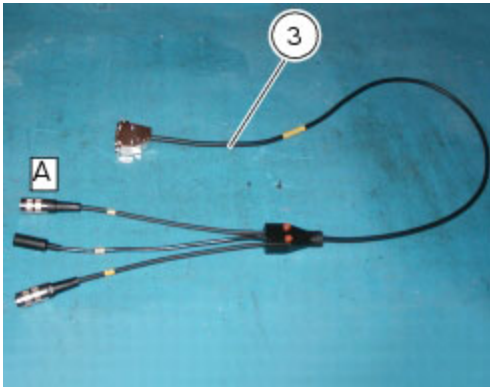


Connect the pressure sensor (5) to socket (A) of the cable (3).

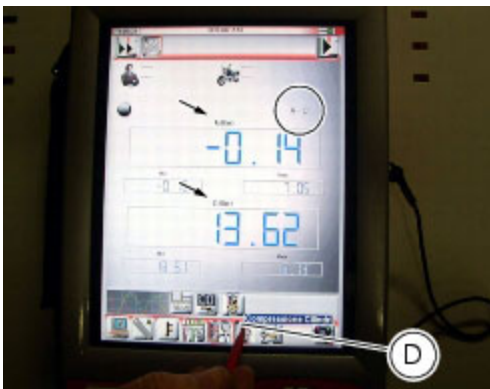


Note

Measure the compression on one cylinder at a time.



On the DDS diagnosis instrument (1), select the "Measurement module" function by pressing the corresponding icon; then press the "Cylinder Compression" icon (D) followed by the "Start" icon (E).



The letter A is displayed on the screen to indicate the connector of the Measurement Module cable (3).

Open the throttles completely.

Turn over the engine with the starter motor until the pressure stops rising.

Check the pressure in each cylinder:

- standard value: **11** to **12** bar;
- minimum value: **10** bar;
- maximum permissible difference between cylinders: **2** bar.



An excessively high pressure value can be caused by:

- build up of deposits in the combustion chamber.

An excessively low pressure value can be caused by:

- gas escaping between the cylinder head and the barrel;
- worn valve seats;
- bent valve stems;
- incorrect valve clearances;
- worn cylinder or piston rings.

Refit the spark plugs and reconnect the HT leads (21).



Fuel pressure test

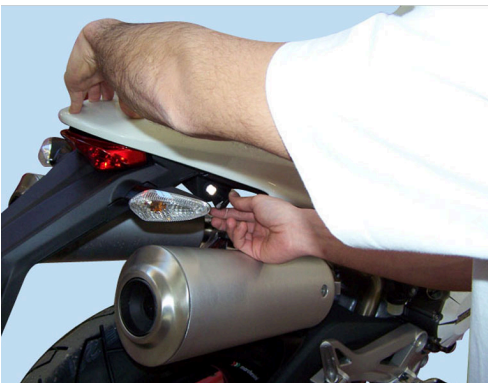


Note

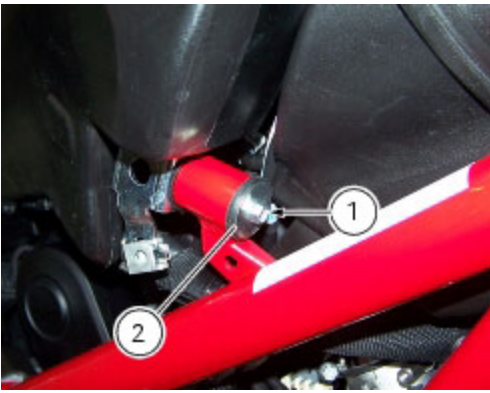
The on-screen icons used during this procedure are explained in a table at the end of this section.

In order to check the valve clearances, it is necessary to gain access to the four valve covers on the cylinder heads and then remove the parts listed below.

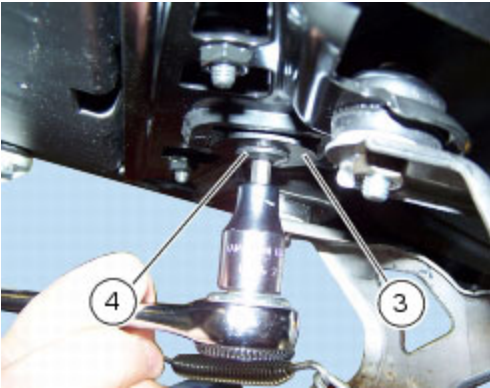
Open the lock and simultaneously apply downward pressure in the area of the catch to release the pin; Pull the seat backwards to release it from the front catches and remove it.



Remove the tank covers (Sect. E 2, [Removal of the fuel tank fairings](#)).
Unscrew the side screw (1) securing tank to the frame and recover the washer (2).



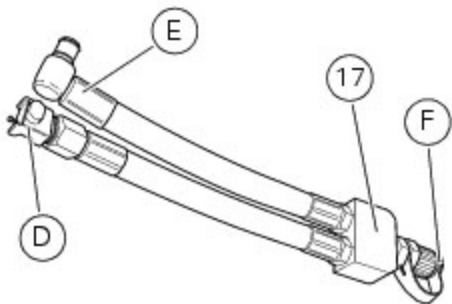
Unscrew the rear screw (3) securing tank and recover the washer (4).



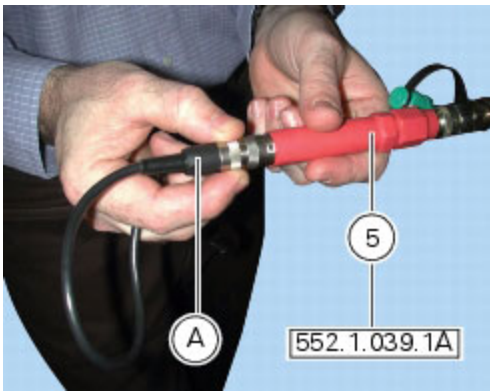
Raise the tank but leave it connected to the fuel system and wiring then position it on the side, duly supported, to be able to work on the fuel system flange.
Remove the fuel feed hose (28).



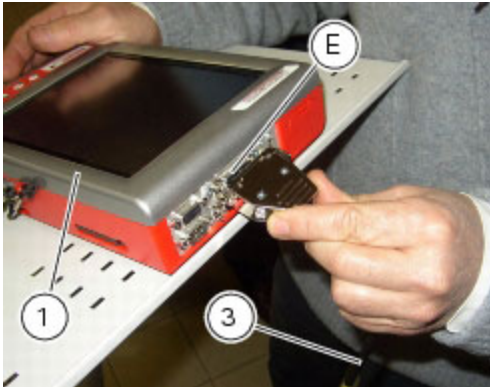
Use the fuel pressure pipe (17) connecting one end (D) to the coupling of the delivery pipe to the tank and the other end (E) to the fuel delivery pipe (28): in this way you create a pressure pick-up socket (F).



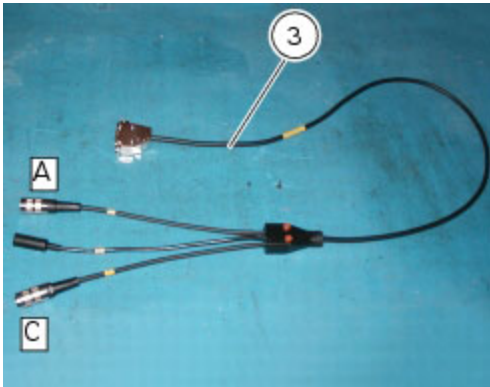
Connect the pressure sensor (5) to the test outlet (F) of the pipe (17) to transform the pressure signal into an electrical signal.



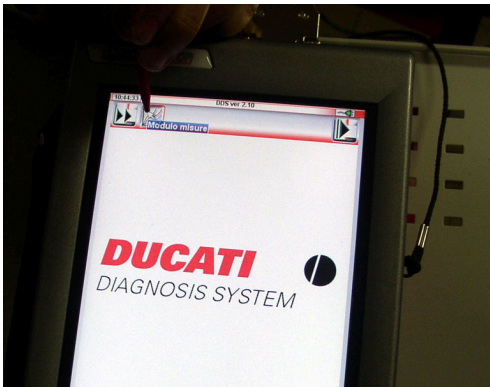
Turn on the DDS diagnosis instrument (1) referring to the paragraph [“Tester power supply”](#).
 Connect the power and diagnostics cable (Measurement Module) (3) to the measurement module connector (E) on the DDS diagnosis instrument (1).



Connect the pressure sensor (5) to socket (A) or (C) of the cable (3).



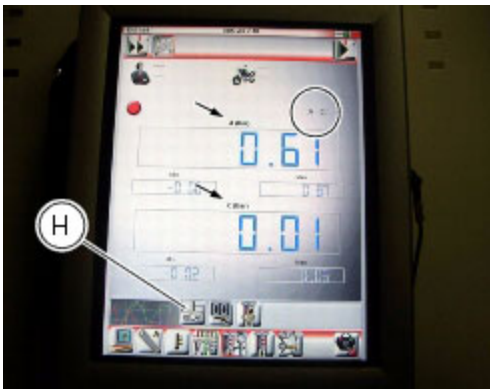
On the DDS diagnosis instrument (1), select the “Measurement module” function by pressing the corresponding icon;
 then press the “Pressure Test” icon (F) followed by the “Start” icon (G).





The socket to which the cable (Measurement Module) (3) is to be connected is indicated on the screen with a capital letter: A, B or C.

The values may be displayed in three different ways: in one numeric form and in two graphic forms; to select the desired display type, press the "Value display" icon (H).



The measured value is indicated alongside the letter (A) or (C) identifying the cable used for the measurement: i.e. if you used connector (A) of the cable (3), the value measured will displayed next to the letter (A) on the screen. The maximum pressure must be equal **3** bar (nominal).

On completion of the test, remove all the components of the test instrument, refit the fuel feed pipe (28) securing it with the clamp.

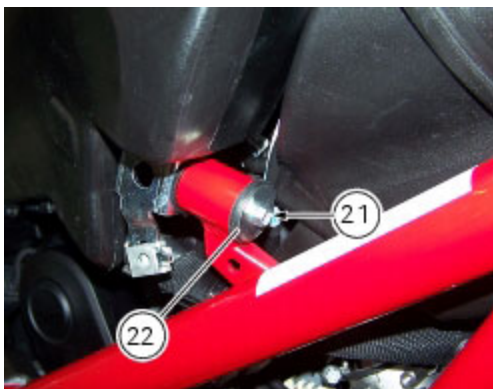


Refit all the removed parts in the reverse order to that in which they were removed.

Refit the tank, ensuring to lay the lambda sensor cable (25) into the relevant tab on the tank flange cover and set the fuel pump connector (26) against the vertical head.



Fix the tank to the frame inserting the screw (21) with washer (22) and tightening it to the specified tightening torque (Sect. C 3, [Frame torque settings](#)).

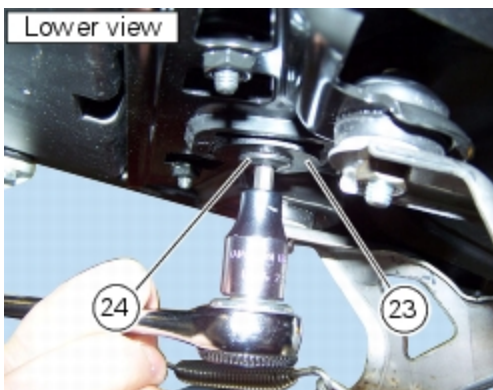


Secure the tank to the subframe with the screw (24) and the washer (23).
Tighten the screw (24) to the specified tightening torque (Sect. C 3, [Frame torque settings](#)) while holding with a size 5 Allen wrench the threaded insert (27) located on the top of the tank.



Important

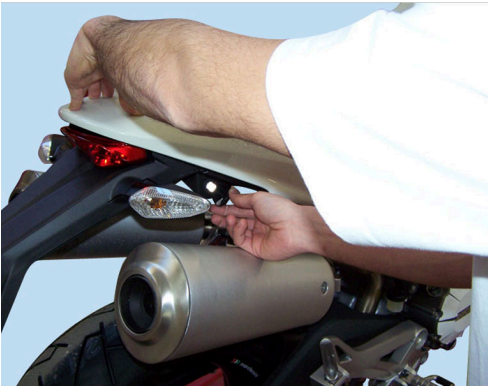
The threaded insert (27) is not present in the USA version.





Refit tank covers (Sect. E 2, [Refitting the fuel tank fairings](#)).

Refit the seat inserting the front ends of the seat base into the tabs on the tank, then push the rear end of the seat until you hear the bolt in the lock click into place.



Immobilizer override procedure



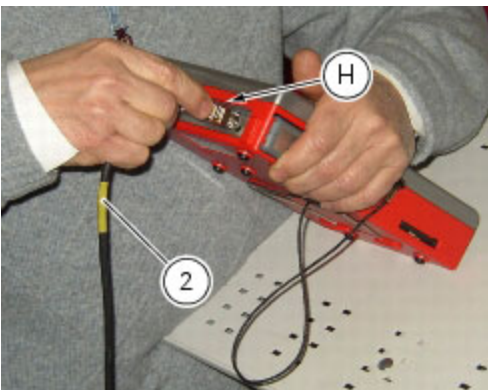
Note

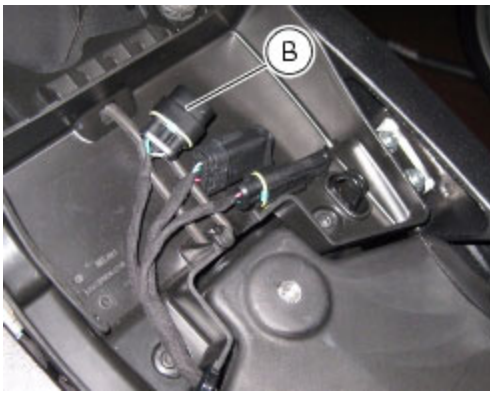
The on-screen icons used during this procedure are explained in a table at the end of this section.

In the event of a malfunction of the immobilizer system, it is possible to start the engine using the emergency procedure described in Section P 8, [immobilizer and transponder](#), which requires a secret code that can be entered using the DDS diagnosis instrument.

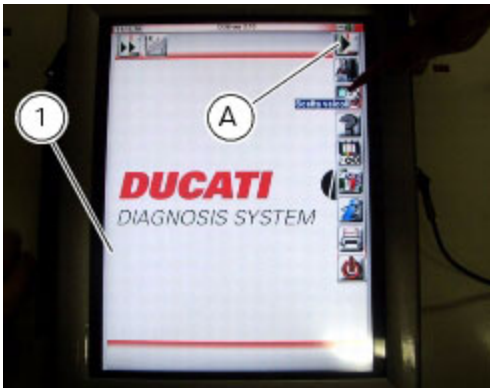
Turn on the DDS diagnosis instrument (1) referring to the paragraph "[Tester power supply](#)".

Connect the power and diagnostics cable (2) to the diagnostics connector (H) on the tester and to the diagnostics socket (L) on the motorcycle.

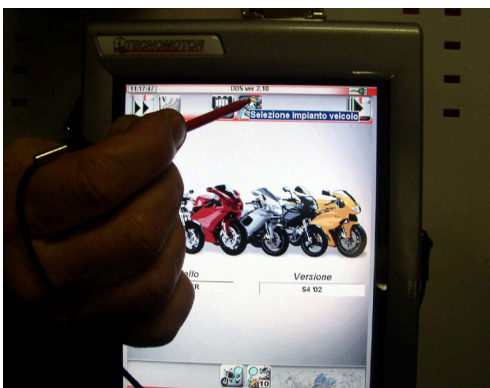




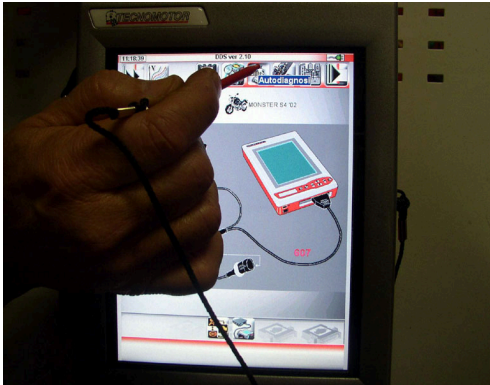
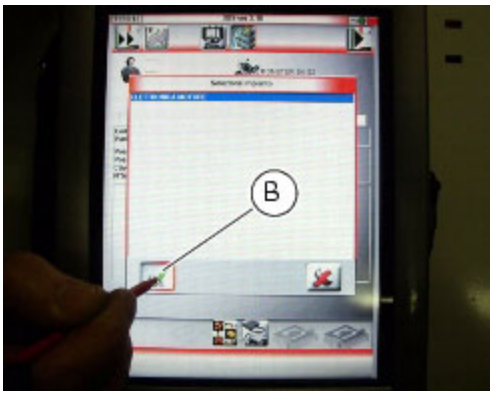
Enter the general functions menu by pressing the icon "Menu 1" (A).



Press the "Select vehicle" icon and, on the next screen, press the "Select motorcycle" icon; select the motorcycle model and confirm, then select the version and confirm.



Press the icon "Select system" to display a list of the bike's systems that can be analysed. Select "Engine electronics" and press the "Confirm" icon (B). Next, press the "Self-diagnosis" icon to access the corresponding function.



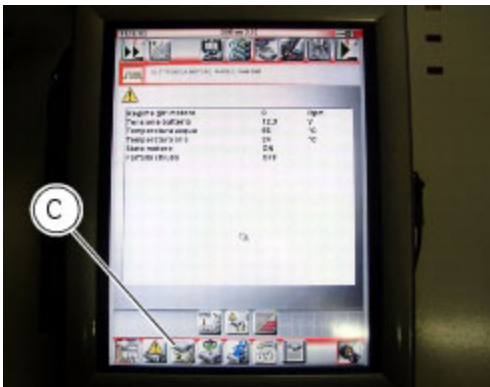
The DDS diagnosis instrument will interrogate the electronic control unit and display the parameters analysed with their relative values.



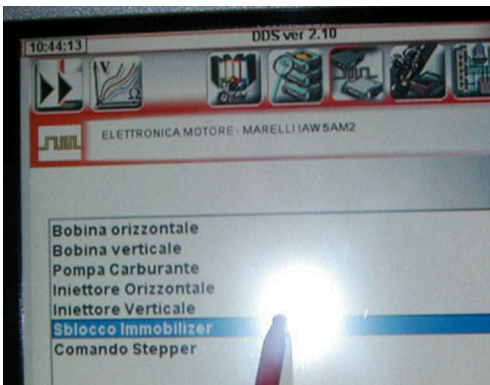
Note

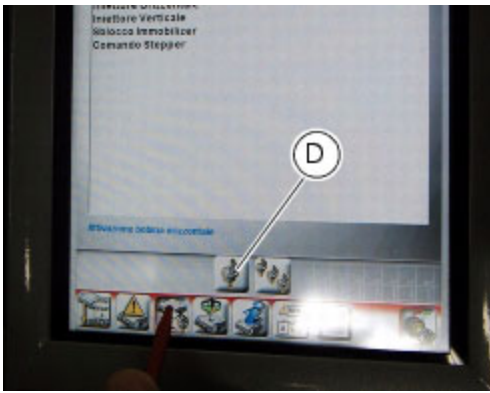
Make all connections with the motorcycle ignition key in the OFF position.

Press the "Actuators" icon (C) to display the actuators and other components connected to the ECU.

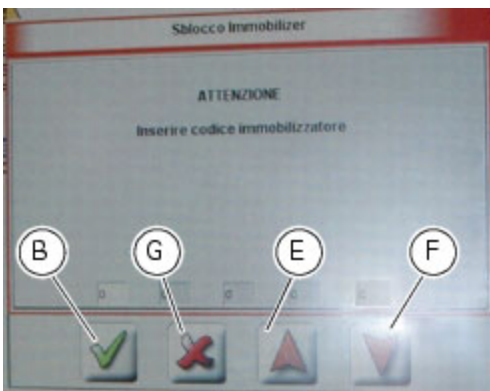


Select the option "immobilizer Override" and then press the icon "Apply" (D).





A screen will appear in which you can enter the 5 digit secret code. Use the red arrows (E) and (F) to enter each digit: after entering each digit, press the "Confirm" icon (B).
 Once the entire code has been entered, press the "Confirm" icon (B) again.
 On completion of the operation, the message "Was the operation completed successfully?" will appear; press "Confirm".
 If any problems were encountered during the operation, the tester will display the relative error messages: each message must be confirmed by pressing the "Confirm" icon, or rejected by pressing the "Exit" icon (G).



Disconnect the tester.

Guided diagnosis



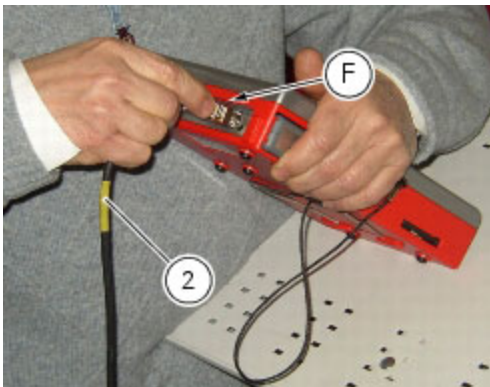
Note

The on-screen icons used during this procedure are explained in a table at the end of this section.

The DDS diagnosis instrument guides the operator step-by-step through the various diagnostic procedures, providing descriptions and documentation for motorcycle components, wiring diagrams for the electronic systems and information on the locations of specific components.

Turn on the DDS diagnosis instrument (1) referring to the paragraph "[Tester power supply](#)".

Connect the power and diagnostics cable (2) to the diagnostics connector (F) on the tester and to the diagnostics socket (G) on the motorcycle.





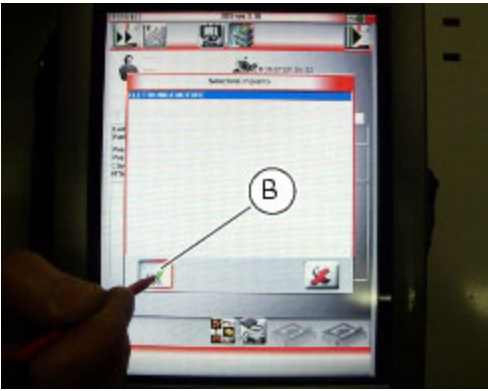
Enter the general functions menu by pressing the icon "Menu 1" (A).



Press the "Select vehicle" icon and, on the next screen, press the "Select motorcycle" icon; select the motorcycle model and confirm, then select the version and confirm.
Press the icon "Select system" to display a list of the bike's systems that can be analysed.



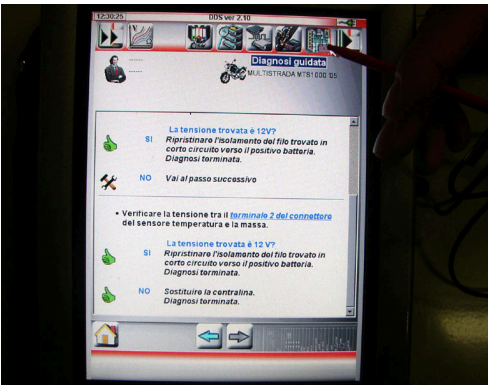
Select "Engine electronics" and press the "Confirm" icon (B).



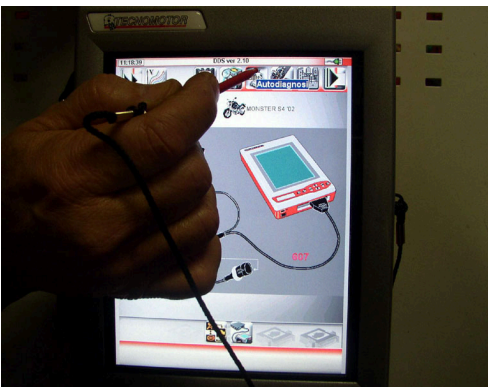
Press the “Guided diagnosis” icon (C) to access the corresponding function.

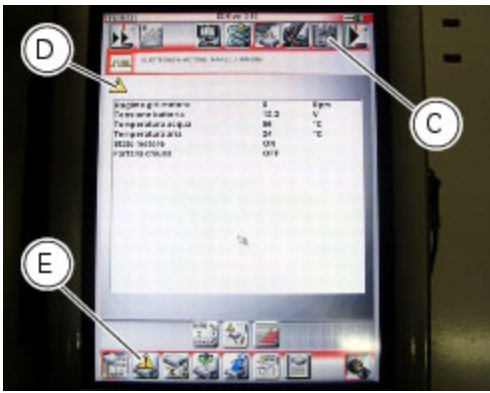


A series of screens are displayed indicating the operations required for correct diagnosis.



To determine whether the system has any internal problems, you can access the “Self-diagnosis” function by pressing the corresponding icon. If any errors are present, the symbol (D) will be displayed. To determine the type of errors present, press the “Errors” icon (E). Once errors have been detected, you can then resolve them through the guided diagnosis procedure; the “Guided Diagnosis” icon (C).





The DDS diagnosis instrument will interrogate the electronic control unit and display the parameters analysed with their relative values.

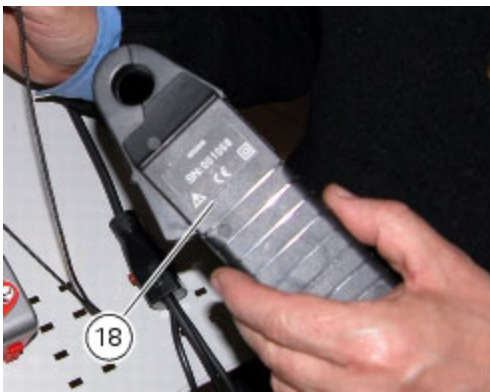
Testing the battery charging system



Note

The on-screen icons used during this procedure are explained in a table at the end of this section.

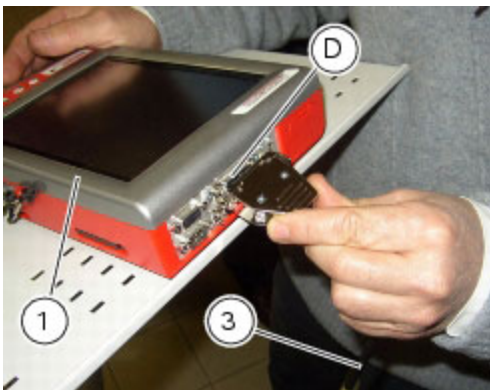
You can determine the engine rpm required for generator to produce just enough current to charge battery, feed the injection ignition system and all electric items fitted to motorcycle. When applied to a cable, the clamp-type amperemeter (18) detects the magnetic field generated by the current passing through that cable.



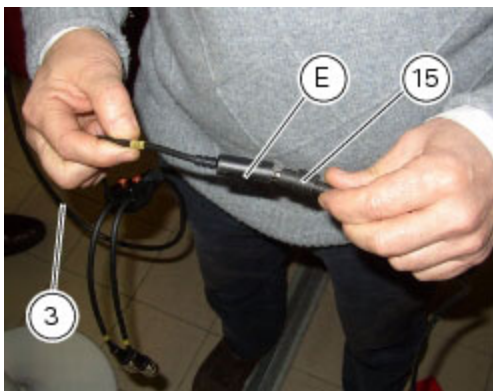
Remove the fuel tank (Sect. L 2, [Removal of the fuel tank](#)).

Turn on the DDS diagnosis instrument (1) referring to the paragraph "[Tester power supply](#)".

Connect the power and diagnostics cable (Measurement Module) (3) to the measurement module connector (D) on the DDS diagnosis instrument (1).



Connect the clamp-type amperemeter to the connector (E) of the power and diagnostics cable (Measurement Module) (3).

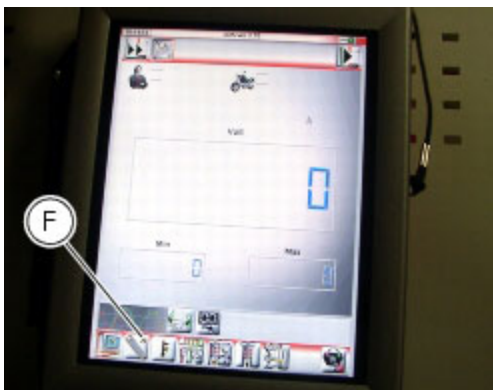
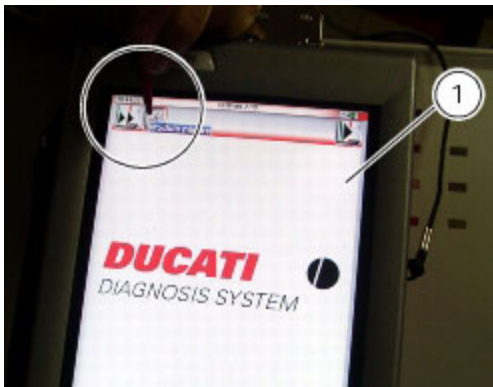


Warning

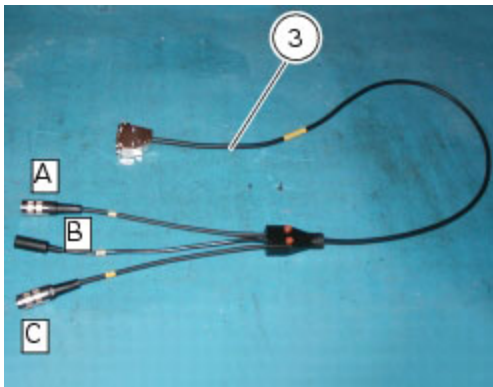
The clamp-type amperemeter must not be connected to wires through which electrical current is flowing.

Insert the clamp-type amperemeter into the battery positive terminal lead with the arrow on the clamp pointing towards the battery positive terminal (+).

On the DDS diagnosis instrument (1), select the "Measurement module" function by pressing the corresponding icon; then press the "Ammeter" icon (F) followed by the "Start" icon.



The socket to which the cable (Measurement Module) (3) is to be connected is indicated on the screen with a capital letter: A, B or C.





If the measured current is a positive quantity, it means that generator is feeding all electric items and charging battery at the same time. If the current has a negative sign, this means that the charging system is not able to power the electrical loads and a significant amount of the current required must be supplied by the battery, which is therefore discharging.



 Important

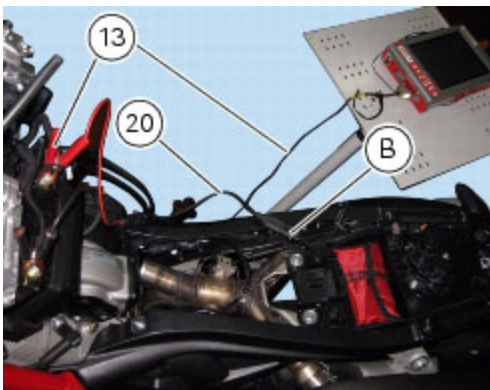
If polarity is reversed when clamping the ammeter onto the cable, the sign of the readings will also be reversed, giving rise to incorrect diagnosis.

ABS diagnosis

 Note

The on-screen icons used during this procedure are explained in a table at the end of this section.

If the ABS system is not working correctly, system diagnosis is possible through the DDS diagnosis instrument. Turn on the DDS diagnosis instrument (1) referring to the paragraph "[Tester power supply](#)": use power supply (13). Connect the power cable (20) to cable (13) and to ABS diagnosis connector (B).



Icons table

Symbol	Description
--------	-------------

	Confirm
	Exit
	Measurement module
	Belt tension
	Start/stop
	Menu 1
	Select vehicle
	Select motorcycle
	Self-diagnosis
	Settings
	Apply
	Cylinder synchronisation
	Value display
	Reset
	Pressure test
	Cylinder compression
	Actuators



Guided diagnosis



Select system



Errors



Clamp-type amperemeter
